



Polycystic ovary syndrome detection using optimized SVM and DenseNet

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Abstract Polycystic ovary syndrome (PCOS) is a complicated endocrine disease that significantly impacts the health of women, affecting fertility and leading to various critical conditions. Unfortunately, around 70% of PCOS cases remain undiagnosed, emphasizing the importance of early detection. Ultrasound imaging has emerged as a valuable tool for detecting polycystic ovaries, providing crucial details such as follicle count, size, and position. However, manual diagnosis through ultrasound imaging is laborious and prone to errors, highlighting the need for more objective diagnostic methods. In this study, we propose two distinct predictive models for PCOS detection, utilizing both text and image based datasets. Firstly, an Optimized Support Vector Machine based PCOS detection model is developed using text-based datasets. Secondly, we introduce an image dataset based PCOS detection model using DenseNet. Experimental results demonstrated the suggested models' effectiveness in accuracy, recall, F-score, and precision for both developed methods. The results showed that the present approaches offer superior performance compared to other methods.

Keywords Polycystic ovary syndrome · PCOS detection · Optimized Support Vector Machine · DenseNet · Classification

1 Introduction

PCOS is one of the most common yet often overlooked endocrine disorders in women today. It is characterized by ovarian dysfunction, marked by a follicle count of ≥ 12 and a size of 2 or 3 mm³. Hyperandrogenism and polycystic ovary morphology (PCOM) are critical defining features of PCOS-related ovarian dysfunction [1]. PCOS is a significant concern in female health, affecting 8–13% of reproductive age women and 6–18% of younger females [2]. Approximately 70% of women with PCOS experience infertility, making it a leading cause of female infertility [3]. While genetic predisposition and geographic location are primary factors contributing to this disorder, lifestyle factors such as an unhealthy diet and infectious diseases can exacerbate the condition. PCOS manifests with various health complications, such as excessive hair growth, weight gain, irregular menstruation cycles, depression, and infertility. In addition to infertility, patients with PCOS often suffer from dysregulation in female hormone levels, elevated levels of male hormones, and hair loss. The syndrome is also associated with serious conditions such as hypertension, psychiatric disorders, cardiovascular morbidity, type 2 diabetes, and other endocrine diseases.

The morphology of the ovaries, that contains several ovarian follicles of varying sizes, is referred to as polycystic [4]. PCOS leads to the formation of multiple sacs in the ovaries. Normally, the ovaries generate follicles, or oocytes, which progress into ova during the hormone-regulated ovulation phase. Hormonal imbalances can disrupt ovulation, causing the continuous production and accumulation of follicles in the ovaries, resulting in the formation of multiple cysts characteristic of polycystic ovaries. In contrast, normal ovaries undergo regular ovulation without such accumulation. Early PCOS identification and diagnosis using minimal testing and imaging processes is critically important due to

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the condition's direct impact on ovarian dysfunction and its associated risks, such as miscarriage, infertility, gynaecological cancer, and the emotional distress and financial burden on patients.

PCOS is diagnosed through three primary methods: blood tests, pelvic exams, and medical imaging. The most prominent technique among these is medical imaging used by healthcare practitioners to identify health conditions. In conventional PCOS diagnosis, practitioners capture a static ultrasound image, position it satisfactorily, and measure the size of the follicles. This process involves manually measuring or marking the follicles to determine if they exhibit PCOS characteristics, repeating these steps for each follicle. However, this manual approach is labor-intensive and susceptible to human error [5]. Automating the diagnosis process can address these issues. Computer-assisted image analysis holds potential for improving the clinical management of PCOS by enhancing the accuracy and efficiency of diagnosis and monitoring. The goal of this study is to aid in the early detection of PCOS, thereby preventing further complications and improving women's health outcomes.

2 Related works

Nilofer et al. [6] used an artificial neural network (ANN) with Improved Fruit Fly Optimization (IFFOA) to classify ovaries in ultrasound images. The images were resized and denoised, followed by follicle segmentation utilizing adaptive k-means clustering. Features were extracted utilizing a statistical gray level co-occurrence matrix (GLCM) and used to train the ANN classifier. The IFFOA-ANN classifier effectively distinguished between normal and abnormal ovaries, aligning well with manual expert detection.

Rachana et al. [7] developed a method to identify PCOS using ultrasound images. The process involved segmentation, feature extraction, and classification with a k-nearest neighbour (KNN) classifier. This method aimed to enhance diagnostic accuracy and speed, reducing the risk of complications from delayed diagnosis. However, it had high computational time for the KNN algorithm.

Shubham Bhosale et al. [8] utilizing a deep convolutional neural network (DCNN) to classify PCOS from ovarian ultrasound images. After removing speckle noise in the pre-processing step, the region of interest was identified. The PCOS segment was segregated using CNN techniques, and features were extracted for classification. The CNN model was trained using training images, and test images were then employed to classify PCOS for model validation. However, an optimal version of CNN was not used to enhance the model.

Homay Danaei Mehr et al. [9] used ensemble classifiers on the Kaggle PCOS dataset for identify PCOS. They tested

various classifiers, including Ensemble Random Forest (RF), Multi-Layer Perceptron (MLP), AdaBoost, and Extra Tree, with both full and reduced feature sets. Feature selection methods improved classifier performance, with Ensemble RF and MLP achieving the greatest accuracy utilizing all features. Ensemble RF, with a reduced feature set, outperformed other classifiers. However, this approach may not be suitable for different datasets.

With reference to ultrasound images, Noraishikin Zulkarnain et al. [10] presented an integration of Otsu's thresholding and the Chan–Vese method to segment and discover follicles in the ovary. Otsu's thresholding brought attention to the image's pixel intensities and created a binary mask that could be employed for establishing follicular boundaries utilizing the Chan–Vese technique. The approach produced better results in the segmentation evaluation. This research demonstrated that the segmentation of ultrasonic images was improved by integrating the Chan–Vese method with Otsu's thresholding. It did not enhance the method's efficacy or applicability for various ultrasound images attributes.

PCONet is a CNN developed by Salman Hosain et al. [11] to detect PCOS in ultrasound images. They fine-tuned an InceptionV3 pretrained model and contrasted it with PCONet, using an independent test set for unbiased evaluation. PCONet proved superior and could aid early PCOS detection in health-care facilities. However, they did not enhance PCONet's accuracy and demonstrate its application to different imaging modalities like photoacoustic and MRI.

Amsy Denny et al. [12] developed an approach for early PCOS identification using key clinical and metabolic parameters. They formulated a feature vector from real-time patient data, identifying eight significant features through PCA. Various ML techniques were tested for classification, with Random Forest Classifier (RFC) proving the most accurate. This automated system aids doctors by reducing diagnosis time and delays in detecting A web-based system that uses machine learning (ML) models for identifying patterns in PCOS was created by Md Mahbubur Rahman et al. [13]. Various algorithms were tested, including Decision Tree (DT), AdaBoost (AB), SVM, RF, and Logistic Regression (LR). In order to evaluate accuracy, specificity, sensitivity, precision, and overall performance, they trained models using data. When selecting features utilizing the Mutual Information (MI) model, AB and RF had the best accuracy.

3 Proposed methodology

3.1 PCOS detection using text dataset

We developed a framework for PCOS detection from a text dataset. Three stages comprise this method: pre-processing, feature selection, and classification. First, data is

acquired and pre-processed, then the crucial features are determined. Feature selection methods such as RFE, L1-regularization, tree-based techniques, and PCA are employed. After that, the chosen features are inputted into our proposed OSVM classifier for categorization as normal or indicative of PCOS. Figure 1 illustrates the model structure. Further details are provided in the subsequent sections.

3.1.1 Pre-processing

Preprocessing is crucial to prepare PCO data for analysis. Initially, features were separated from the target variable, followed by normalization. Exploratory Data Analysis (EDA) helped understand PCOS diagnosis distribution and feature-target relationships, identifying significant predictors. Various methods, including data augmentation, were employed to enhance dataset diversity and model adaptability.

3.1.2 Feature selection

Following pre-processing, feature selection is conducted to identify the main attributes of the data. The goal is to build a subset of significant attributes that improve classifier performance by reducing overfitting and enhancing accuracy. Methods such as RFE [14], L1 Regularization [15], Tree-based techniques [16], and PCA [17] are employed to extract features. Ten important features were selected from pool of 45, including Follicle No. (R), Follicle No. (L), Skin darkening (Y/N), Hair growth (Y/N), Weight gain (Y/N), Cycle (R/I), Fast food (Y/N), Pimples (Y/N), AMH (ng/mL), and Weight (Kg).

3.1.3 Classification using OSVM

After feature selection, the selected features are sent into optimized SVM (OSVM) classifier to classify a text data as normal or PCOS. In this study, we propose an OSVM

classifier, a supervised learning model, for classifying PCOS. The OSVM leverages the robust capabilities of SVM, a technique introduced by Vapnik, designed to solve function approximation problems through statistical learning theory. Over the past two decades, SVMs have become a popular tool in pattern recognition, with applications extending beyond classification to include regression tasks. In a predetermined amount of time, SVM converges to a global solution since it resolves a convex quadratic problem. SVMs are useful for many different types of applications, including text classification in a range of domains. The utilization of kernel-induced feature spaces, sparseness of the solution, and lack of local minima are the main features of SVM.

The objective is to create a model that forecasts the target values of the test data using the attributes of the test data. SVM attempts to construct a hyperplane with the greatest possible margin between classes. SVM are designed to divide a set of training images into two distinct classes, represented as $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where x_i in R^d , the d-dimensional feature space, and y_i in $\{-1, +1\}$, the class label, with $i = 1, \dots, n$. SVM constructs the optimal separating hyperplanes using a kernel function (K). All data points, whose feature vectors lie on one side of the hyperplane, are assigned to class -1 , while those on the other side belong to class $+1$ [18]. As illustrated in Fig. 2, H3 fails to separate the two classes, while H1 separates them with a narrow margin. Only H2 provides the maximum margin between the two classes, making it the correct hyperplane utilized by the support vector machine. If the data from different classes can be separated as shown in Fig. 2, a linear SVM is used. Conversely, if the data cannot be separated, a non-linear SVM classifier is employed. Instead of defining a function for the hyperplane itself, the margin between the two classes is defined. Figure 3 shows that the position of our hyperplane depends on the value of w , where w are the (not necessarily normalized) normal

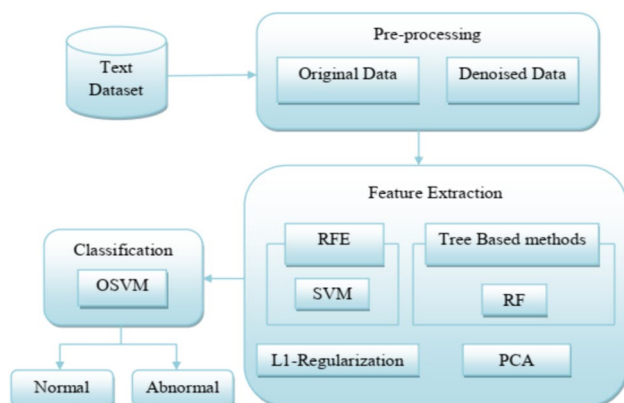


Figure1 PCOS detection using OSVM

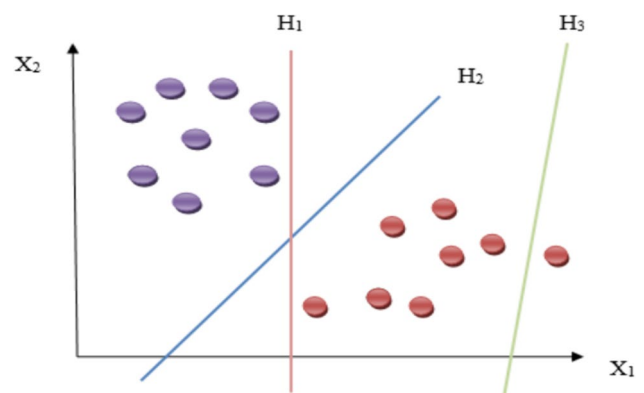


Fig. 2 Representation of separating hyperplanes for two classes

vector to the hyperplane. More formally, the linear SVM classifier function can be defined as follows:

$$f(x) = w^T x + b \quad (1)$$

In this way, the function returns $f(x_i) > 0$ for $y_i = -1$ and $f(x_i) < 0$ for $y_i = +1$ for every training sample x_i . Stated otherwise, the hyperplane $f(x) = w^T x + b = 0$ divides training samples of two distinct classes, with w representing the weight vector and normal to the hyperplane, b denoting the bias or threshold, and x_i representing the data point. Here is the definition of the nonlinear SVM classifier

$$f(x) = w^T \Phi(x) + b \quad (2)$$

The primary challenge is that a linear hyperplane often fails to accurately separate the data points of binary classes, resulting in higher classification errors. To address this, by utilizing kernel functions, SVM is able to convert the data into a feature space of higher dimension, allowing for the linear separation of classes. A kernel function is defined as $k(x_i, x_j) = \Phi(x_i)^T \Phi(x_j)$, where Φ is the feature vector, on two samples given as feature vectors in some input space. The most often utilized kernels are the sigmoid, polynomial, RBF, and linear types. The choice of model, which includes the kernel function and its parameters, is the primary determinant of SVM performance. In this work, we select the polynomial kernel defined by the Eq. (3):

$$k(x_i, x_j) = (\gamma X_i^T X_j + r)^d \quad (3)$$

where d is the polynomial's degree and r is a free parameter that trades off the influence of the polynomial's higher-order vs lower-order elements, T is the transposition of a matrix,

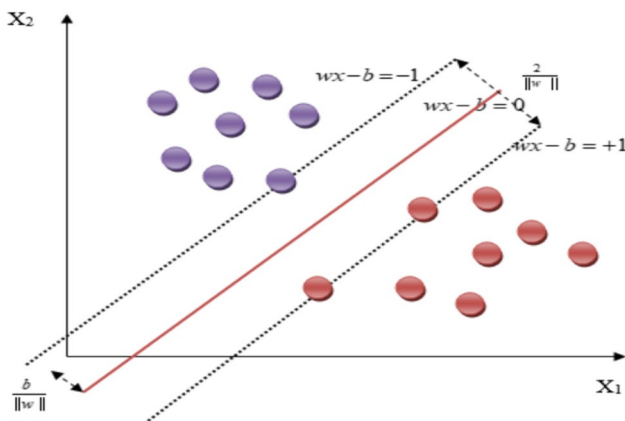


Fig. 3 Maximum-margin hyperplane and margins for SVM training with samples from two classes

and γ and r are parameters. Where γ and r are parameters optimized to improve the SVM classification performance. To enhance the SVM model's performance, we employ the Walrus Optimization Algorithm (WaOA), inspired by walrus behavior, particularly their strategies in feeding, migration, and predator evasion [19]. WaOA optimizes the SVM model's core parameters of r and γ to minimize prediction error rates and improve classification accuracy. The entire parameter optimization process using WaOA can be explained as below.

Step 1: Initialization Randomly initialize the walrus populations represented by a solution C vector. This population is represented by the C vector.

$$C = \{c_1, c_2, c_3, \dots, c_N\} \quad (4)$$

Step 2: Fitness function The WaOA evaluates the fitness of each solution based on classification accuracy. The solution with the highest accuracy represents the best fitness.

$$\text{Fitness} = \text{Max}(\text{Accuracy}) \quad (5)$$

Step 3: Updation using WO After evaluating fitness, solutions are updated using the AWO update function, mirroring the natural behaviors of walruses across three phases.

Phase 1: Feeding strategy (Exploration) The strongest walrus leads the group in foraging. Similarly, in the WO algorithm, the best solution is guided by the walrus with the highest objective function value. This feeding behavior improves the algorithm's exploration by updating walrus positions using Eq. (6).

$$c_{ij}^{P_1} = c_{ij} + \text{rand}_{ij} \cdot (WX_j - I_{ij} \cdot c_{ij}) \quad (6)$$

$$C_i = \begin{cases} C_i^{P_1}, & F_i^{P_1} < F_{i,*} \\ C_i, & \text{else,} \end{cases} \quad (7)$$

where $F_i^{P_1}$ is the fitness function value, rand_{ij} are random values from the interval $[0,1]$, $C_{ij}^{P_1}$ is the j th dimension, WX is the best candidate solution—which is thought to be the strongest walrus—and $C_i^{P_1}$ is the newly produced location for the i th walrus based on the beginning phase, and I_{ij} is utilized to enhance the algorithm's capacity for exploration, resulting in more important and wide-ranging changes in the walruses' position when set to 2, as opposed to the value of 1, which represents the displacement's default state.

Phase 2: Migration The WaOA Algorithm utilizes walruses' natural migration patterns, to explore the search space. Each walrus moves to a new, randomly selected position. According to Eq. (9), the new position replaces the original location if it increases the objective function value.

$$C_{ij}^{P_2} = \begin{cases} c_{ij} + \text{rand}_{i,j} \cdot (c_{m,j} - I_{i,j} \cdot c_{ij}), & F_m < F_i, \\ c_{ij} + \text{rand}_{i,j} \cdot (c_{ij} - c_{m,j}), & \text{else}, \end{cases} \quad (8)$$

$$C_i = \begin{cases} C_i^{P_2}, & F_i^{P_2} < F_i; \\ C_i, & \text{else}, \end{cases} \quad (9)$$

In the second phase, $C_i^{P_2}$ represents the new produced location of the i -th walrus, $C_{ij}^{P_2}$ is the j th dimension, $F_i^{P_2}$ is the fitness function value, C_m , $m \in \{1, 2, \dots, N\}$ and $m \neq i$, indicates the position of the chosen walrus to move the i -th walrus towards it, $s_{m,j}$ denotes its j -th dimension, and F_m is the value of its fitness function.

Phase 3: Fighting off and escaping from predators (exploitation) Walruses constantly face threats from predators like polar bears and killer whales. To mimic their natural behavior of evading and confronting these threats, the WAOA algorithm adjusts walrus positions near their current locations. This enhances the algorithm's local search capabilities around candidate solutions. Initially, the neighbourhood radius is set to its maximum to prioritize global exploration, then gradually decreases to focus on local optimization. Local lower and upper bounds ensure a variable radius with each iteration. Equations (10) and (11) randomly generate new positions within this neighbourhood, and these positions replace the previous ones if they improve the objective function value, as per Eq. (12).

$$C_{ij}^{P_3} = c_{ij} + (la_{local,j}^t + (ua_{local,j}^t - \text{rand} \cdot la_{local,j}^t)) \times LF \quad (10)$$

$$\text{Local bounds} : \begin{cases} la_{local,j}^t = \frac{la_j}{t}, \\ ua_{local,j}^t = \frac{ua_j}{t}, \end{cases} \quad (11)$$

$$C_i = \begin{cases} C_i^{P_3}, & F_i^{P_3} < F_i; \\ C_i, & \text{else}, \end{cases} \quad (12)$$

where $C_i^{P_3}$ is the newly produced position for the i th walrus based on the third phase, $C_{ij}^{P_3}$ is its j th dimension, $F_i^{P_3}$ is its fitness function value, t is the iteration contour, la_j and ua_j are local lower and higher bounds that are permissible for this j th variable, respectively, to simulate local search in the neighbourhood of the candidate solutions.

Step 4: Termination The algorithm iterates through the above steps until the optimal values of γ and r are identified.

By employing WAOA to optimize SVM parameters, our proposed OSVM model effectively classifies PCOS with improved accuracy and minimized prediction error rates.

3.2 PCOS detection using image dataset

We introduce a framework for PCOS detection using image data. Our methodology consists of three stages: pre-processing, feature extraction, and classification. First, data is gathered and pre-processed. After pre-processing, the image dataset's important features are extracted and used to classify images using a DenseNet method, which indicates whether PCOS is present or absent. Figure 4 outlines our model structure. Detailed methodology explanation follows.

3.2.1 Pre-processing

The image dataset preprocessing involved augmenting the data through techniques like rotation, flipping, and zooming, resulting in five additional images per original image. This expanded dataset was then resized to a standardized dimension of 256×256 pixels with three color channels for consistency in input dimensions. This meticulously managed dataset, enhanced through data augmentation, provides a valuable asset for training and evaluating methods designed to detect and classify polycystic ovaries in ultrasound images. Below is the mathematical representation of the image pixel transformation, where Π_q denotes the scale of the images.

$$\Pi_{(p,q,r)} = \sum_{j=0}^n \Pi_{q(256,256,3)} \quad (13)$$

Normalization was applied to convert the ultrasound images into numerical representations, ensuring uniformity for mathematical operations. For efficient model training and assessment, the dataset was further split into training, testing, and validation sets in the ratios of 70:10:20.

3.2.2 DenseNet-based feature extraction and classification

After preprocessing, the input images are inputted into the DenseNet model for the purpose of feature extraction and classification. DenseNet is used to classify a data as normal

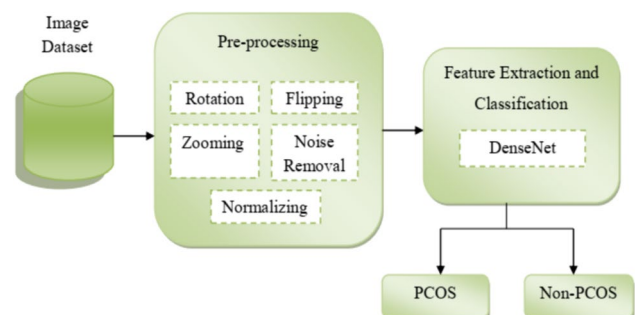


Fig. 4 PCOS classification using DenseNet

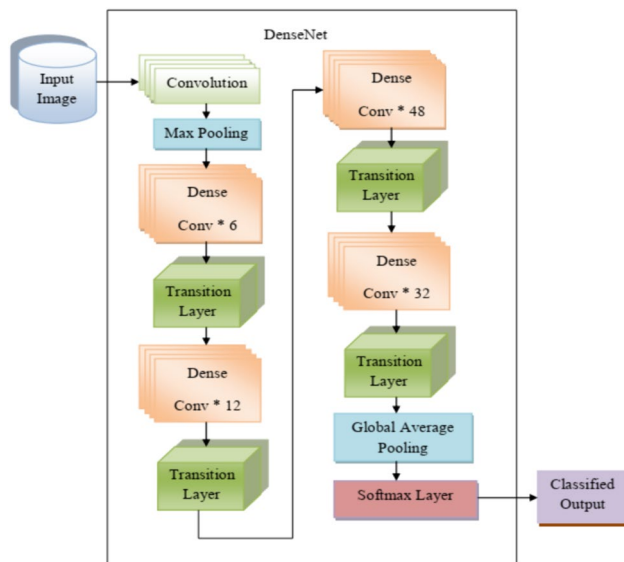


Fig. 5 DenseNet method flow

or PCOS. Figure 5 depicts the suggested DenseNet structure's flow. The pre-processed images are fed straight into the first convolution (Conv) layer of the model to train it. CNNs learn features related to the task and are highly effective at extracting unique characteristics that thoroughly represent the image. Through automatic learning, they acquire features at different abstraction levels, allowing them to transform raw input data into the desired output by learning complex functions. In our suggested model, complete features are retrieved from the image datasets using DenseNet, and DenseNet is then trained using these features. Fine details can be extracted from the images by each dense layer.

An initial convolutional layer, a max-pooling layer, four dense convolution (Dense Conv) blocks, and four transition layers— 1×1 Conv and 2×2 average pooling—were used to construct the suggested method. A set of 1×1 Conv and 3×3 Conv blocks make up dense convolution blocks. These convolutions (1×1 and 3×3) are repeated 6, 12, 48, and 32 times within each Dense Conv block following each alternate transition layer. After going through these layers, the output feature maps are provided as an input for the Global Average Pooling (GAP) block. GAP is followed by a fully connected (FC) layer. The data samples are categorized into the appropriate classes by the FC layer.

3.2.2.1 DenseNet DenseNet, proposed by Huang in 2017, is a deep network architecture designed to address network degradation and enhance the use of features across layers. DenseNet is a further innovation of ResNet. DenseNet offering a simpler yet more effective structure with fewer parameters compared to traditional CNN. This architecture is particularly beneficial in reducing overfitting problems

that often arise with smaller training sets, such as those used in PCOS classification. DenseNet develops a full connected module known as Dense Block, which consists of multiple fully connected layers (Dense layers). Every layer in DenseNet [20] is directly connected to every other layer, allowing for efficient information transfer between them. Every layer receives additional inputs from all preceding layers and shares its feature maps with all subsequent layers. The current layer's output feature maps are merged with those of the previous layer through concatenation. Each layer is connected to all of the network's subsequent layers; these networks are known as DenseNets. Figure 6 depicts the dense layer's structure.

Imagine that the convolutional network processes an input image, y_0 . Each of the N layers in the network performs a nonlinear transformation $F_n(\cdot)$. Assume that all feature maps from previous convolutional layers make up layer n . Layers 0 to $n - 1$ input feature maps are concatenated and shown as $y_0, \dots, y_{n-1}$. As a result, the N -layer network in this model contains $N(N + 1)/2$ connections. The n -th layer's output is provided by

$$y_n = F_n([y_0, \dots, y_{n-1}]) \quad (14)$$

where $F_n(\cdot)$ the composite function of Batch Normalization (BN) is followed by Rectified Linear Units (ReLU), y_n is the current n -th layer, and $[y_0, \dots, y_{n-1}]$ is a concatenation of feature maps acquired from 0 to $n - 1$ layer. For managing the feature map dimensions, a transition layer is added after each Dense Block. The transition layer includes operations such as BN, ReLU, and 3×3 Conv. When feature maps sizes change, concatenation operation is not feasible, therefore, downsampling is performed using 1×1 Conv and 2×2 average pooling.

7×7 Convolution blocks with a stride of 2 make up the first convolution layer. Global average pooling and the softmax classifier comprise the classification layer that comes after the final Dense Convolution block. All of the neural network's feature maps are used to make the accurate prediction. The K -neuron output layer corresponds to the detection of PCOS-related conditions. Convolution preserves the pixel

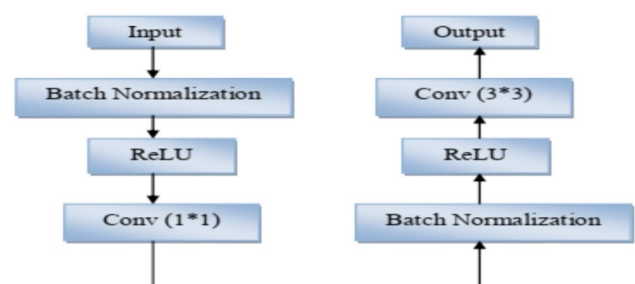


Fig. 6 Dense layer architecture

connections while learning the characteristics of the image. Every layer of convolution relates to the BN-ReLU-Conv sequence. ReLU is applied to the output feature maps following the convolution of the image. CNNs become nonlinear as a result of this function. Equation (15) provides the ReLU function.

$$\text{ReLU}(y_0) = \max(0, y_0) \quad (15)$$

The process of pooling is used to lower the dimensionality of the output feature maps. Either max pooling or average pooling is used for this pooling. Taking the largest component from the enhanced feature map is the process of max pooling. By dividing the input into the pooling area and computing the average values of each area, average pooling is performed. Every feature map is averaged by GAP, and the resultant vector is sent to the softmax layer. One fully connected softmax layer makes up the categorization layer. Multi-class classification problems are categorized using the softmax function. The probability distributions of every class i over all possible classes are computed by this function. The activation function of softmax is provided by.

$$s(z_p) = \frac{e^{z_p}}{\sum_j e^{z_q}} \quad (16)$$

where z_p is the input value and z_q is the all input values of I . The formula determines the ratio between the input elements's exponential and the total of all the input data's exponential values. Hence, at the end of the output layer, both PCOS and non-PCOS are classified. DenseNet innovative architecture and efficient feature utilization make it an excellent method for PCOS classification. By leveraging dense connections and transition layers, DenseNet not only reduces the risk of overfitting but also ensures robust and accurate classification performance.

4 Results and discussion

The study outcomes were generated using a Google Colab notebook, utilizing computing resources featuring an Intel i3 7th generation 2-core processor at 2.30 GHz and 8 GB of RAM.

4.1 Dataset description

4.1.1 Text dataset

The original dataset comprised 781 images per category. In the textual dataset, 3550 instances were analyzed, encompassing 45 feature columns such as 'Skin darkening (Yes/No)', 'Menstrual cycle irregularity (Regular/Irregular)', 'Consumption of fast food (Yes/No)', 'Presence of pimples (Yes/No)', 'Weight gain (Yes/No)', 'Hair growth (Yes/No)', and others.

4.1.2 Image dataset

The Image dataset comprises ultrasound images categorized into two groups based on the presence or absence of polycystic ovaries. Label 1 denotes images showing uteruses with polycystic ovaries, distinguished by bilaterally enlarged ovaries, numerous tiny follicles, and heightened stromal echogenicity [21]. Conversely, Label 0 indicates images lacking these characteristics. Sample representations from the dataset are depicted in Fig. 7, where the left corresponds to Label 1 indicating PCOS presence, while the right corresponds to Label 0 indicating absence of PCOS.

4.2 Results of a comparative analysis

In this section, we contrast our suggested approach with other methods. Our system comprises two primary methods: text dataset-based PCOS detection using OSVM and image dataset-based PCOS detection using DenseNet. We assess each stage independently, evaluating performance with metrics including accuracy, precision, recall, and F1-score.

4.2.1 Results of OSVM method based PCOS detection

Figure 8 presents an analysis of OSVM-based PCOS detection performance utilizing the text-based dataset, compared with the varying classifiers such as SVM, KNN, and GNB. According to Fig. 8, the suggested approach attained a maximum accuracy of 97.5%, surpassing SVM-based PCOS detection by 2.11%, KNN-based detection by 15.5%, and GNB-based detection by 1.65%. Similarly, our suggested strategy achieved a highest precision of 90%, recall of 93%, and F-measure of 85%. Figure 8 demonstrates that in terms of results, the suggested approach performed better than the current approaches.

4.2.2 Results of DenseNet based PCOS detection

Our suggested DenseNet's effectiveness evaluation is contrasted with that of InceptionV3, CNN, and SqueezeNet in Fig. 9. As illustrated in Fig. 9, the suggested approach

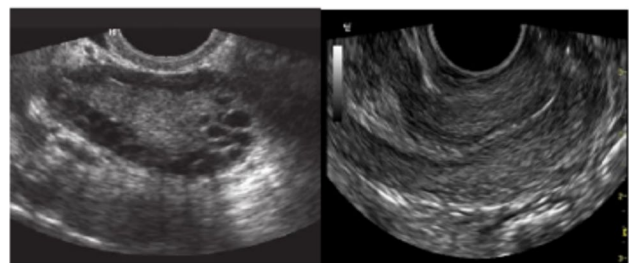


Fig. 7 Representations of classes within the image dataset

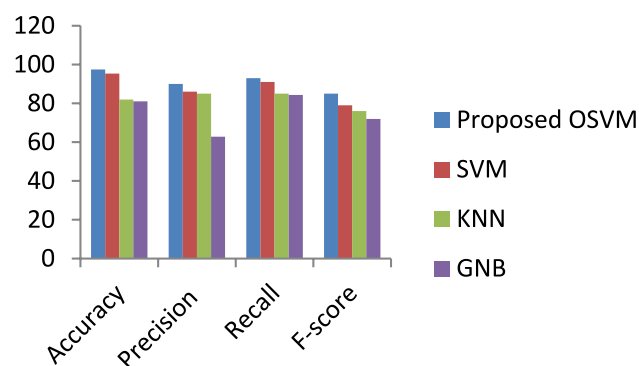


Fig. 8 Comparison of OSVM with existing methods

yielded the best accuracy of 97.68%, surpassing InceptionV3-based PCOS detection by 1.12%, CNN-based detection by 12.68%, and SqueezeNet-based detection by 22.14%. Similarly, our suggested strategy achieved the maximum precision of 95.2%, recall of 93.57%, and F-measure of 96.04%. In terms of results, Fig. 9 demonstrates that the proposed approach performed better than the current approaches.

4.2.3 Comparison with published studies

To demonstrate the efficacy of our suggested method, we compared this study with previously published studies. Table 1 presents the results of this comparative analysis, focusing on various PCOS detection methods. Detailed descriptions of each work are provided in the literature survey.

According to Table 1, our proposed method achieved the highest accuracy: 97.5% for the text-based dataset and 97.68% for the image-based dataset. In contrast, the accuracies reported in studies [6, 7], and [9–13] were 96.5%, 97%, 96.15%, 89.93%, 96.2%, 89.02%, and 94%, accordingly. Our method attained the highest accuracy, surpassing

Table 1 Results of a comparative analysis

References	Accuracy	Precision	Recall	F-Measure
[6]	96.5	86	91	84
[7]	97	–	–	–
[9]	96.15	87.5	–	84.96
[10]	89.93	–	–	–
[11]	96.12	89	92	83
[12]	89.02	88.83	–	41.82
[13]	94	89.94	88.57	84
<i>Proposed</i>				
Text-based	97.5	90	93	85
Image-based	97.68	95.2	93.57	96.04

current methods. Furthermore, proposed technique achieved a maximum precision of 90% in the text-based dataset and 95.2% in the image-based dataset, recall of 93% in the text-based dataset and 93.57% in the image-based dataset, and an F-score of 85% in the text-based dataset and 96.04% in the image-based dataset. These results consistently demonstrate the superiority of our proposed technique in accurately detecting PCOS.

5 Conclusion

One of the most prevalent endocrine disorders affecting women who are fertile is PCOS, which often leads to anovulation and infertility. In this work, we developed two distinct methods to classify PCOS based on text and image datasets. Three steps were involved in the execution of the proposed approaches: preprocessing, feature extraction, and classification. For the text-based dataset, data were first collected and then underwent preprocessing. The most optimal features were selected using methods such as RFE, L1-Regularization, tree-based methods, and PCA. These selected features were then classified as PCOS or non-PCOS using the OSVM classifier, achieving an impressive accuracy of 97.5%. For the image-based dataset, data were also gathered and pre-processed. The DenseNet method was employed to select the most optimal features and classify the data as PCOS or non-PCOS, yielding an exceptional accuracy of 97.68%. The effectiveness of the proposed strategies was comprehensively analyzed using various metrics and compared to several PSOS detection techniques. The proposed strategies demonstrated superior performance compared to other existing methods. Future research could focus on developing a real-time diagnostic tool that combines text and image data for a holistic PCOS detection approach. Collaboration with clinical experts would be essential to validate and refine the models for practical use.

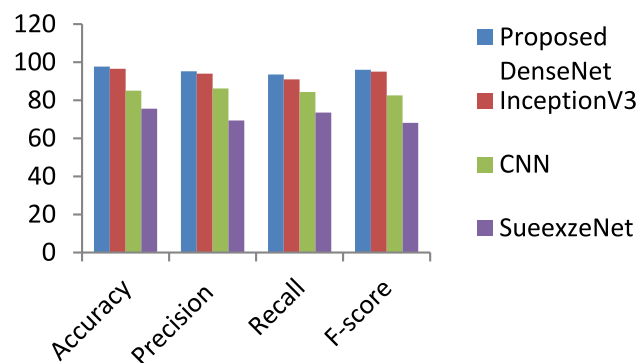


Fig. 9 Comparison of DenseNet with existing methods

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Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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